

Effect of Local Maternity Ward Closures on Health and Demographic Outcomes

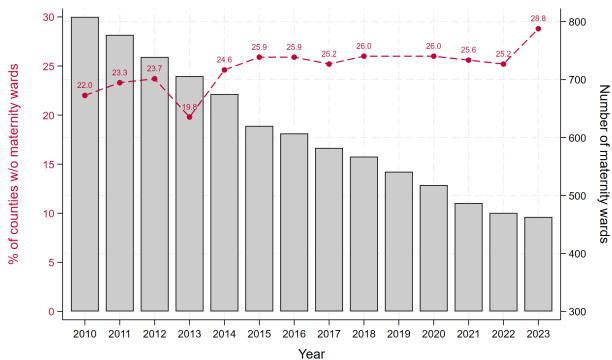
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Motivation

- The number of obstetric hospitals and obstetricians has decreased, leading to a continuous reduction in delivery infrastructure (Ministry of Health and Welfare, 2012–2024).
 - mainly due to a sharp decline in birth rates, low delivery fees, and the burden of medical malpractice
- The number of institutions handling deliveries decreased by 40.4% over 10 years, from 777 in 2011 to 463 in 2023.
- The South Korean government established a support program for obstetric care in underserved geographic areas (OCUA) but numbers of counties without maternity wards have consistently been increasing.

Motivation



Notes. Data from Ministry of Health and Welfare (2012–2024).

Motivation

- To be designated as an underserved area, a county must meet the following criteria:
 - ① less than 30% of pregnant women in the county can access a delivery room within one hour
 - ② more than 30% of women of childbearing age in the county cannot access a delivery room within one hour
- Most studies on the accessibility of maternity wards for South Korean mothers and its related health implications have focused on comparisons between OCUA and non-OCUA regions (Kwak et al., 2019b; Kim et al., 2024), but these criteria can create blind spots.
- None of the counties that lost maternity wards were able to re-establish maternity care facilities without government support.
- It is essential to conduct a comprehensive empirical study that considers the demographic characteristics of mothers residing in these regions, even if they are outside of OCUA areas.

Literature

In this study, I aim to investigate the impact of the closure of maternity wards on delivery methods and maternal and infant health outcomes.

- Lack of nearby maternity wards limits the options available to mothers for their preferred delivery method.
 - Lewis et al. (2019) denoted that the closure of these facilities often results in a rise in early elective deliveries, including inductions and C-sections, as women prefer not to risk going into labor far from the nearest hospital.
 - Park et al. (2021) found that in 2021, the probability of undergoing a C-section was 47.3% in urban neighborhoods (Dong), compared to 59.5% in rural towns and townships (Eup, Myeon). discrepancy
 - This discrepancy may partly be explained by the greater distances to maternity wards in rural regions (Kim et al., 2023).
- The extended travel times to maternity units and inadequate access to obstetric care are also known to be correlated with increased neonatal mortality and other negative health outcomes for both the mother and the child (Ravelli et al., 2011; Grzybowski et al., 2011; Kwak et al., 2019a).

Data

- 2002-2022 administrative records from the National Health Insurance System (NHIS) for all women who had a confirmed pregnancy at least once during the period from 2012 to 2022
 - ① Eligibility database:
Patients' age, county of residence, insurance type, monthly insurance fee, date of death
 - ② Medical record database:
medical treatments, including diagnoses of diseases, date of visit, healthcare provider identifiers, medical treatments, medical expenses
 - ③ Healthcare provider database:
location of hospitals, the medical specialties offered, number of healthcare professionals
- I extracted a sample of mothers who gave birth using relevant procedure codes and found total of 3,235,611 birth records.
- Each mother and child are linked using unique individual identifiers.

Variables	ICD-10 codes
A. Health conditions related to high-risk pregnancies	
Pre-labor	O60.0, O60.1, O60.2, O60.3
Bleeding	O67.0, O67.8, O67.9, O72.0, O72.1, O72.2, O72.3
Eclampsia	O11, O13, O14, O15, O16
Premature rupture of membranes	O42
Premature separation of placenta	O45
Placenta previa	O44, O69.4
Polyhydramnios	O40
Oligohydramnios	O41.0
Antepartum haemorrhage	O46
Cervical incompetence	O34.3
Hypertension	O10, O13, O16
Multi-fetus	O30, O31
Diabetes mellitus	O24, O24.4, O24.9, E10, E11, E12, E13, E14, E15
Hyperemesis gravidarum with metabolic disturbance	O21.1
Kidney-related diseases	N00-N23
Heart-related diseases	I00-I52
Poor fetal growth	O36.5
Other diseases with cervix organs	O23.5, O34.0, O34.1, O34.4, O34.8, O41.1
Rupture of uterus	O71.0, O71.1
B. Health conditions that necessitates C-section	
Maternal care for disproportion	O33.9
Failure of induction	O61
Abnormal labor force	O62
Prolonged labor	O63
Obstructed labor	O64, O65, O66
Complicated labor delivery	O68, O69
C. Postpartum maternal health conditions	
Puerperal infections	O85, O86, O86.0, O86.1, O86.2, O86.3, O86.4, O86.8
Postpartum complications	O75, O87, O88, O90
Perineal laceration	O70
Miscarriage and abortions	O02.1, O03, O04, O05, O06
D. Postpartum infant health conditions	
Low birth weight (< 2500g)	P07.0, P07.1
Extreme immaturity (Gestation < 28 weeks)	P07.2
Preterm birth (Gestation 28-37 weeks)	P07.3

Method

I employ the following TWFE specification for difference-in-difference (DID) estimation:

$$Y_{ct} = \beta Treat_c \times Closed_{ct} + \delta X_{ct} + \gamma_c + \gamma_t + \gamma_{ct} + \varepsilon_{ct}$$

- $Treat_c$: indicator equal to one for the closure counties
- $Closed_{ct}$: indicator equal to one following the closure of the last remaining maternity ward
- X_{ct} : time-varying, county-level controls
 - Maternal characteristics (e.g., average age of birth, average insurance fee)
 - Other demographic statistics (e.g., employment rates by gender, population density, and female ratio)
 - pre-birth maternal health indicators
- For β to estimate an unbiased treatment effect in DID settings, the parallel trends assumption must hold.

Method

- A key concern is that mothers who are living in an area with better availability for maternity wards may differ in characteristics such as underlying health conditions and economic status (Hadley and Cunningham, 2004; Card et al., 2023).
- This study address the problem in two ways:
 - 1 I replicate method from Fischer et al. (2024) by predicting each county's propensity to experience closure (p) and using $\frac{p}{(1-p)}$ as the weight to give weight to non-closure counties similar to the closure counties. regression
 - 2 I utilize event study specification to examine whether the observed outcomes align with the timing of the ward closures.

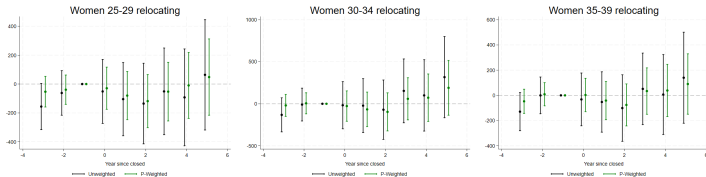
$$Y_{ct} = \sum_{\tau \neq -1} \beta_{\tau} (Treat_c \times 1\{t_c^* + \tau = t\}) + \delta X_{ct} + \gamma_c + \gamma_t + \gamma_{ct} + \varepsilon_{ct}$$

- Following a staggered DID framework, the treated group consists of 5 counties all of which experienced the closure of maternity wards during the sample period, and control group consists of 136 counties where maternity wards remain open throughout the sample period.

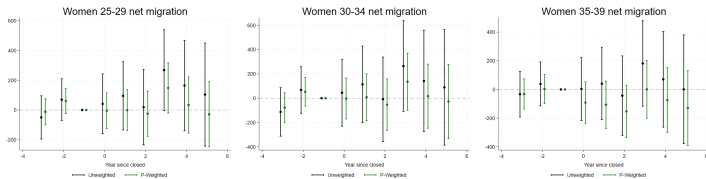
	Closure Counties	Non-Closure Counties Unweighted	Non-Closure Counties P-Weighted
Total Number of Birth	732.1273 (658.6439)	2343.805 (1885.287)	628.8448 (605.7816)
Population	100867.2 (53567.7)	329130 (224621.7)	97258.16 (62897.74)
Population Density	375.2114 (463.485)	5962.905 (6862.495)	1166.075 (2732.618)
Female Employment Rate	51.46 (5.6242)	49.7314 (4.7831)	51.4088 (4.9079)
Male Employment Rate	72.4764 (3.8342)	70.1622 (4.0335)	70.3309 (4.1054)
Female 20-24 Population Share	2.3217 (0.3779)	3.0249 (0.4694)	2.551 (0.4516)
Female 25-29 Population Share	2.0069 (0.4272)	2.9712 (0.7743)	2.1344 (0.5696)
Female 30-34 Population Share	2.4156 (0.8468)	3.1956 (0.8068)	2.4299 (0.6996)
Female 35-39 Population Share	2.9385 (0.9423)	3.5505 (0.6699)	2.8804 (0.6632)
Married Female 20-24 Population Share	0.2254 (0.1535)	0.1262 (0.1089)	0.1567 (0.1436)
Married Female 25-29 Population Share	0.7533 (0.2728)	0.6983 (0.268)	0.7268 (0.2384)
Married Female 30-34 Population Share	1.8504 (0.8079)	1.9474 (0.6049)	1.691 (0.5927)
Married Female 35-39 Population Share	2.4311 (0.9664)	2.7125 (0.6321)	2.3569 (0.5929)
Insurance Fee for Mothers (KRW)	87145.28 (14202.32)	92667.98 (15676.77)	85092.94 (13351.0)
N of Counties	5	136	36

Demography Check I

(a) Relocation

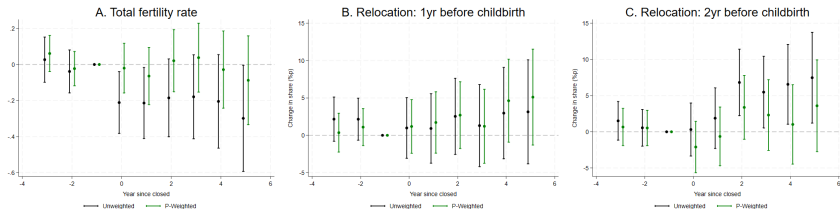


(b) Net Migration



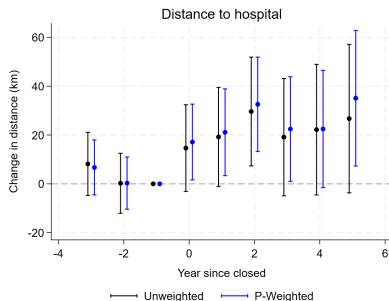
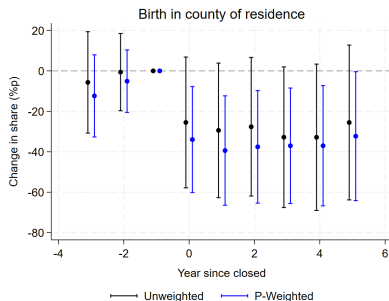
Demography Check II

(c) Total Fertility Rate and Relocation of Mothers



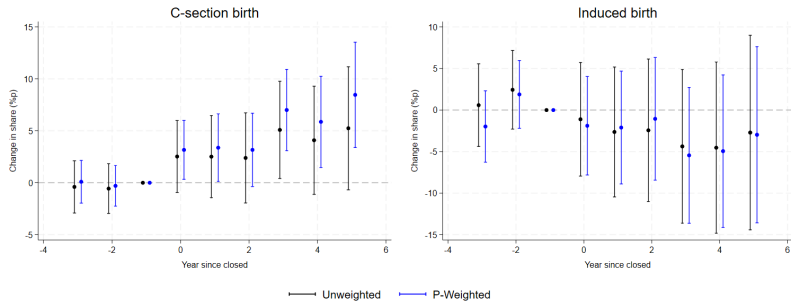
- There is no significant change in the out-migration and net migration of women of childbearing age (aged 25-39).
- The fertility rate shows a significant decline after closure of maternity wards when comparing to the entire 136 control groups.

Mother's Location of Births



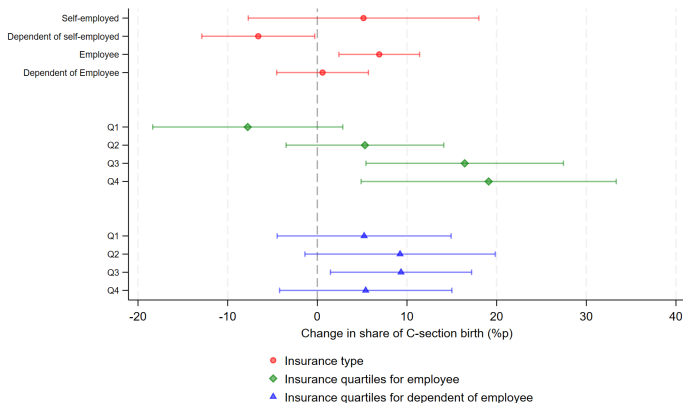
- There was significant decrease of 43.93%p ($p\text{-value} < 0.001$) in births within the counties where mothers resided after closure.
- The average travel distance increased by 21.89 km ($p\text{-value} = 0.003$), as mothers were compelled to seek maternity care in neighboring counties.

Result: Birth Method



- C-section birth increased by 5.43%p (p-value = 0.021) after the closures, while there was no significant increase in induced birth.

C-section by Insurance Type

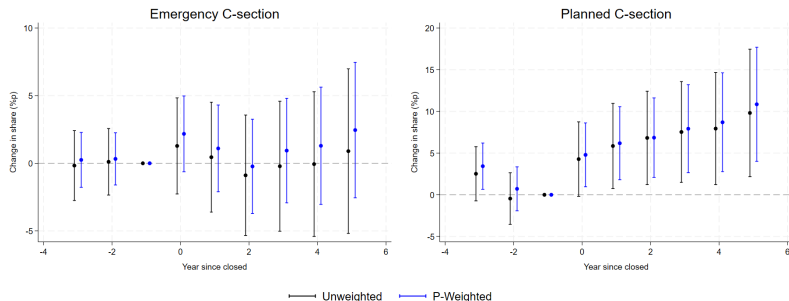


- C-section birth rate increased most significantly among employed mothers when categorized by insurance type (6.91%p, p-value = 0.003).
- When stratified into quartiles based on their insurance fees, the increase in C-sections was particularly evident in the higher quartiles, suggesting that mothers with less flexibility in their work schedules were more likely to be affected by the closures.

C-section by Cesarean Need Tercile

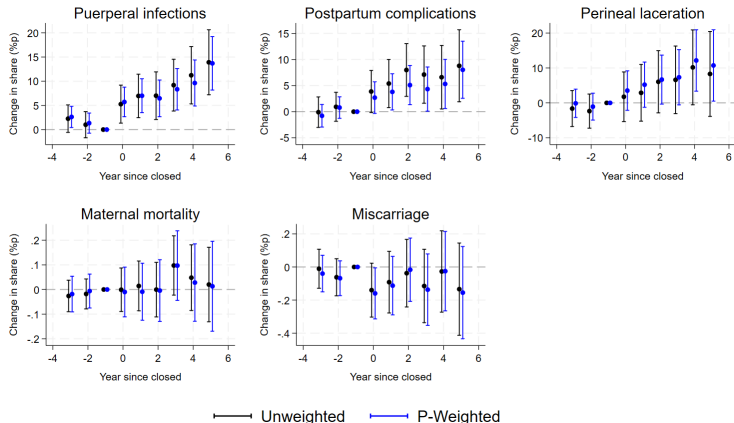
Cesarean Need Tercile	C - section Birth			Induced Birth		
	0-33	33-66	66-100	0-33	33-66	66-100
Closed	5.930*** (2.279)	3.677 (4.354)	-0.549 (2.465)	-6.818 (6.113)	3.768 (4.720)	-2.306 (1.594)
Prior C-section	179.7 (124.2)	214.1** (103.2)	47.58*** (4.016)	-286.5 (333.3)	117.4 (111.9)	-14.78*** (2.597)
Previous child count	-12.66*** (2.139)	32.51** (13.84)	-19.23*** (3.666)	9.031 (5.737)	-32.05** (15.00)	-0.0927 (2.371)
Employment Status and Insurance Fee of Employed Mothers						
Employed mothers (%)	9.211 (7.859)	18.24 (11.58)	-30.75*** (7.323)	85.95*** (21.08)	0.987 (12.55)	6.086 (4.736)
Employed mothers 1yr pre-birth (%)	13.27* (7.666)	15.82 (12.15)	27.58*** (7.157)	-71.39*** (20.56)	-32.17** (13.16)	-4.606 (4.628)
Insurance fee (10,000 KRW)	-1.814** (0.881)	2.325* (1.403)	0.613 (0.754)	-1.794 (2.362)	-3.002** (1.521)	1.091** (0.487)
Insurance fee 1yr pre-birth (10,000 KRW)	-0.173 (0.954)	-4.140** (1.671)	-2.898*** (0.947)	-0.00396 (2.559)	6.659*** (1.811)	0.865 (0.613)
Hospital Characteristics (Base: Births in Tertiary Hospital)						
Births in general hospital (%)	-16.72** (8.135)	-49.50*** (11.50)	-10.61** (4.268)	32.15 (21.82)	26.46** (12.47)	4.377 (2.760)
Births in hospital (%)	-11.54 (8.416)	-39.72*** (12.18)	-5.051 (4.486)	28.34 (22.58)	34.24*** (13.20)	7.635*** (2.901)
Births in clinic (%)	-18.08** (8.353)	-59.04*** (12.15)	-4.661 (4.402)	28.39 (22.41)	37.50*** (13.16)	1.960 (2.846)
Constant	32.75*** (9.370)	34.38 (21.62)	103.0*** (8.062)	48.62* (25.14)	87.10*** (23.43)	-2.334 (5.213)
Mean Dep. Var. (t-1)	9.19%	27.86%	94.07%	39.74%	31.64%	2.42%
N	382	382	382	382	382	382

Emergency and Planned C-section



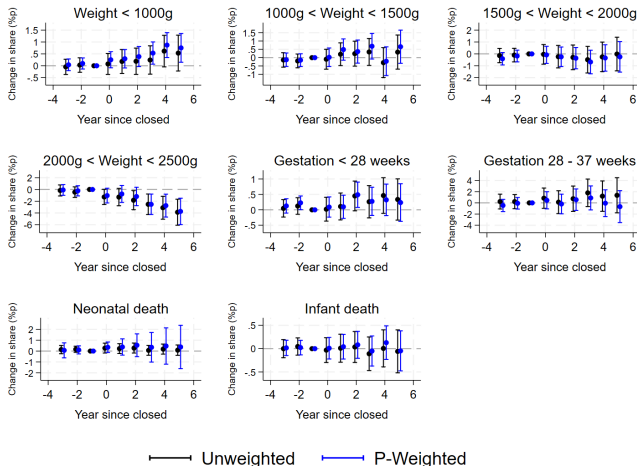
- The rate of emergency C-sections due to immediate safety concerns remained unchanged, the proportion of planned C-sections rose.
- Post-closure, mothers with low medical need for a C-section may have opted for the procedure to mitigate the increased travel burden.

Result: Maternal Health



- The rates of puerperal infections and postpartum complications rose by 3.34%p and 2.56%p, respectively, with a similar upward trend observed in perineal lacerations.
- There were no significant impact of these closures on more severe outcomes, such as maternal mortality or miscarriage.

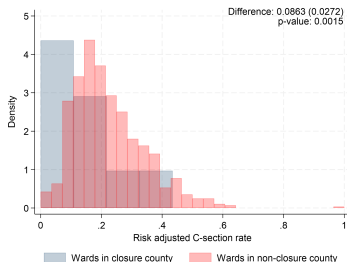
Result: Infant Health



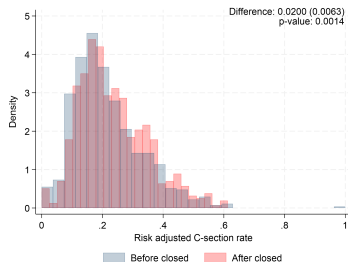
- While there was no statistically significant rise in low birth weight ($< 2,500\text{g}$), a more detailed breakdown shows an increase in the number of births of infants weighing less than $1,000\text{g}$ ($0.33\% \text{p}$, $p\text{-value} = 0.054$).

Result: C-section Practices at Hospitals

(a) Closure and Non-closure Counties



(b) Before and After Wards Closure



Notes. Risk adjusted C-section Rate for each hospital = $(\text{actual \# of C-sections}) / (\text{predicted \# of C-sections}) \times \text{C-section Rate}$

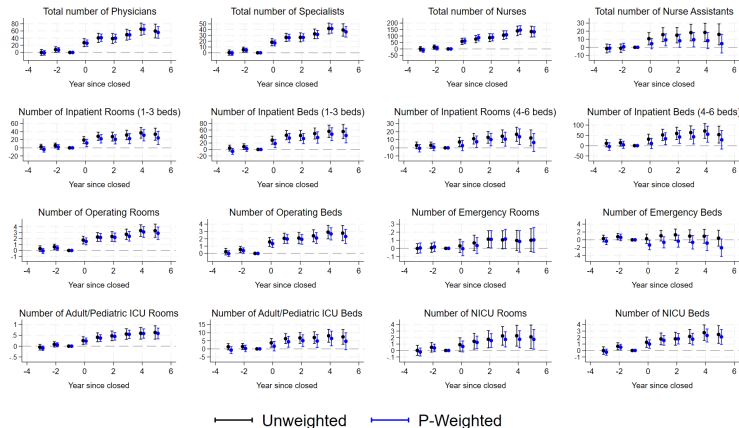
- Maternity wards in closure counties had an average C-section rate of 13.77%, which was 8.63%p lower (p-value = 0.0015) than the rate observed in maternity wards outside of closure counties.
- The C-section rate of the maternity wards mothers used in the post-closure period was, on average, 2%p higher (p-value = 0.0014) than that used in the pre-closure period.

Mechanism

The increase in C-section rates and adverse health outcomes may due to:

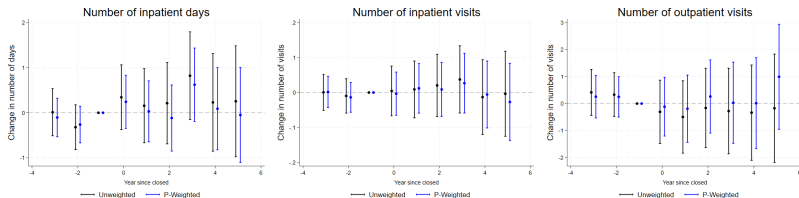
- ① Lower hospital quality:
 - Previous studies highlighted that the learning-by-doing mechanism in hospitals improves the appropriateness of healthcare service (Kristensen et al., 2014; Avdic et al., 2019)
 - Bertoli and Grembi (2019) found that lower-quality hospitals are more likely to overuse C-sections.
- ② Reduced prenatal care:
 - Reduced hospital accessibility may decrease the frequency of prenatal check-ups, potentially contributing to worsening health outcomes.
- ③ Scheduled birth due to increased burden of travel distance:
 - As mothers do not want to risk going into labor far from the hospital, they often choose early elective delivery methods (Lewis et al., 2019).

Mechanism: Change in Hospital Quality



- Hospitals mothers relocated had higher number of physicians, specialists, nurses, and nurse assistants.
- They had more NICU (Neonatal Intensive Care Unit), a feature strongly associated with lower mortality rates for low-birth-weight infants (Phibbs et al., 2007).

Mechanism: Change in Prenatal Visits



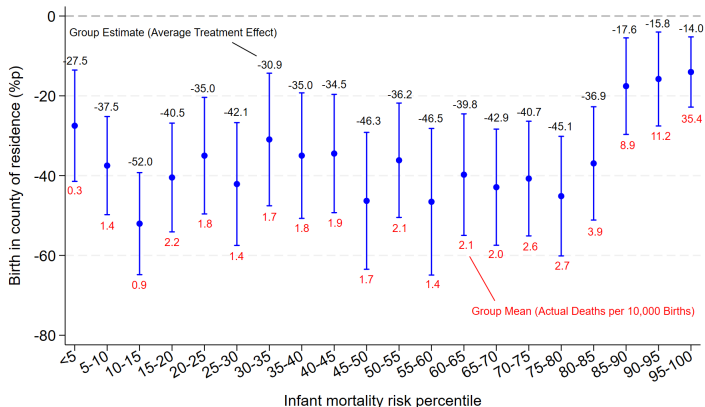
- No statistically significant changes were observed in the length of hospital stays or the number of visits.
- Although maternity wards were closed in the affected counties, obstetric services for prenatal check-ups remained available across all regions.

Mechanism: Distance-based Effect of Closures

	(1)	(2)	(3)	(4)	(5)	(6)
	C-section	Planned C-section	Puerperal Infections	Postpartum Complications	Perineal Laceration	Maternal Mortality
Distance (km)	0.0376*** (0.0105)	0.0256** (0.0124)	0.00296 (0.0120)	0.0429*** (0.0109)	0.0411** (0.0206)	0.000341 (0.000372)
Constant	35.71*** (13.76)	15.43*** (11.67)	-3.493 (15.75)	-47.26*** (14.30)	-22.83 (27.03)	-0.823* (0.488)
N	382	382	382	382	382	382
Pseudo R^2	0.982	0.8223	0.858	0.944	0.805	0.353

- Every additional kilometer in average distance increase the likelihood of C-section by 0.04%p (p-value < 0.001), and planned C-section by 0.026 (p-value = 0.041).
- Postpartum complications and perineal lacerations also rose by 0.04%p (p-value < 0.001, p-value = 0.094), respectively, with each kilometer.

Heterogeneity: Location of Birth by Predicted Mortality Risk



- Mothers above 85% of the distribution were already more likely to give birth in external counties before the closures, leading to a smaller post-closure decrease in the likelihood of giving birth locally, compared to lower-risk mothers.

Robustness Check: Propensity Score Matching

- To address potential aggregation bias and correlated regression errors that may arise from a county-level grouped structure (Moulton, 1986), I apply propensity matching using individual-level data to check the robustness of the results.
- The treatment group consists of mothers residing in closure counties who gave birth after the closure of maternity wards, while the control group includes mothers from counties that never experienced closures.
- For the matching process, I apply nearest-neighbor matching, selecting the three closest neighbors based on propensity scores.
- A total of 15,071 treated individuals were matched each with three individuals from the control group.

Robustness Check: Propensity Score Matching

(%)	(1) All counties	(2) <10km	(3) 10 - 20km	(4) 20 - 30km
Birth Method				
C-section	-0.1586 (0.5252)	-1.847*** (0.6303)	4.4021*** (1.4301)	4.091*** (1.1078)
Planned C-section	-1.7238*** (0.3324)	-3.6163*** (0.3891)	2.6574*** (0.9657)	4.3519*** (0.786)
Maternal Health				
Puerperal infections	-1.0959*** (0.1373)	-1.3256*** (0.1615)	-1.6646*** (0.3114)	0.3766 (0.3507)
Postpartum complications	-5.3376*** (0.2352)	-6.5386*** (0.2702)	-4.7112*** (0.6977)	0.4963 (0.618)
Perineal laceration	-1.4312*** (0.2832)	-2.1361*** (0.3288)	-3.7252*** (0.7559)	3.2957*** (0.714)

- For mothers residing within 10 km of a maternity ward, the likelihood of undergoing a C-section decreases by 1.85%p, with modest improvements in maternal health outcomes.
- Mothers residing 20-30 km from the nearest maternity ward experience a 4.09%p increase in C-section likelihood, along with a 3.3%p increase in perineal lacerations.

Conclusion

- Reduction in maternity wards and subsequent decline in maternity ward accessibility contributing to the increased rate of C-sections and worsening health outcomes for mothers.
- These effects are particularly pronounced for mothers living in counties where the distance to the nearest maternity ward increased significantly following closures.
- Recently, various efforts such as establishing maternity waiting homes and government subsidies for maternity wards have emerged to address disparities in maternal healthcare accessibility.
- Even counties not designated as underserved areas still face significant disadvantages following maternity ward closures.
- To ensure safe deliveries for women, particularly those who face distress due to their distance from delivery facilities, the need for establishing a nationwide maternal healthcare delivery system should be emphasized.

Thank you!

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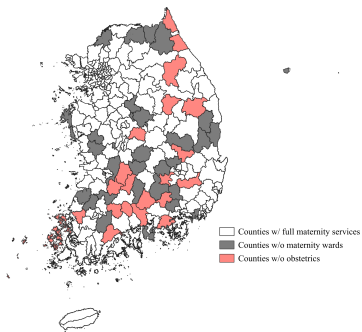
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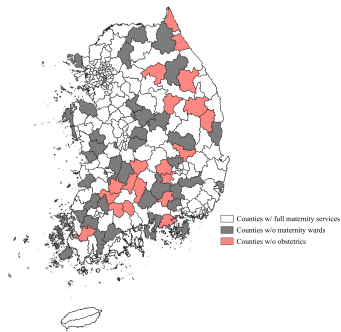
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Counties without Obstetrics or Maternity Wards

(a) 2012

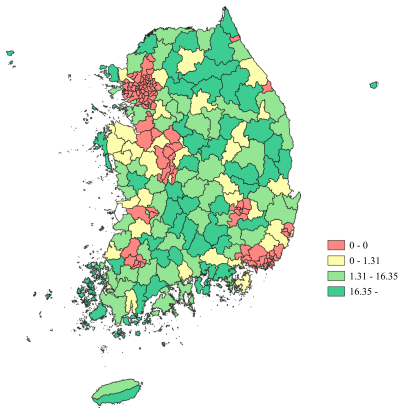


(b) 2022



Notes. Data from Ministry of Health and Welfare (2012–2024). The figures depict the counties without obstetric services and maternity wards for each year.

Discrepancy of the Accessibility of Maternity Wards



Notes. Data from National Medical Center (2022). This figure represents the proportion of women of childbearing age (age 15–49) who cannot access maternity wards within 60 minutes, calculated as the number of such women divided by the total female population of childbearing age.

Closure Logit Regression Estimates

Total Fertility Rate	-1.954 (1.883)
Total pop. (per 1,000,000)	-11.43* (6.029)
Pop. density (population/ km^2)	-0.000226* (0.000133)
Female employment rate	-0.0513 (0.0665)
Male employment rate	-0.00184 (0.0751)
Female pop. share 15-19	-299.2* (181.7)
Female pop. share 20-24	59.52 (166.5)
Female pop. share 25-29	-95.37 (170.7)
Female pop. share 30-34	301.5 (227.2)
Female pop. share 35-39	-410.7 (298.6)
Female pop. share 40-44	267.3 (265.4)
Female pop. share 45-49	-54.13 (205.3)
Insurance fee of employed mothers	-6.716 (16.55)
Insurance fee of employed mothers 1yr pre-birth	-7.358 (17.40)
Constant	16.89** (8.503)
N	168
Pseudo R^2	0.4489